

Medicine-Time Reminder

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We are all so busy with our daily lives that we forget to take our medicines on time, which can have several ill effects on our health. The circuit described here helps in making a simple reminder that allows you to set an alert for the medicines that you need to take. The device flashes an LED or rings a buzzer at the selected time intervals. The available time intervals are matched with standard dosing times such as 4, 6, 8, 12, 24 and 48 hours for your convenience. A perfect



TABLE I
Time Period Selection

DIP1 and DIP2 status	Time interval in hours
Pins 6 and 7 short	4
Pins 5 and 8 short	6
Pins 4 and 9 short	8
Pins 3 and 10 short	12
Pins 2 and 11 short	24
Pins 1 and 12 short	48

TABLE II

Test Points

Test point	Details
TP0	GND
TP1	9V
TP2	2Hz
TP3	A pulse every 30 minutes

time-interval makes this device a better choice amongst other standard reminders.

Circuit and working

The circuit diagram of the reminder is shown in Fig. 1. The circuit is built around 14-stage ripple-carry binary counter CD4060 (IC1), 12-stage ripple-carry binary counter CD4040 (IC2),

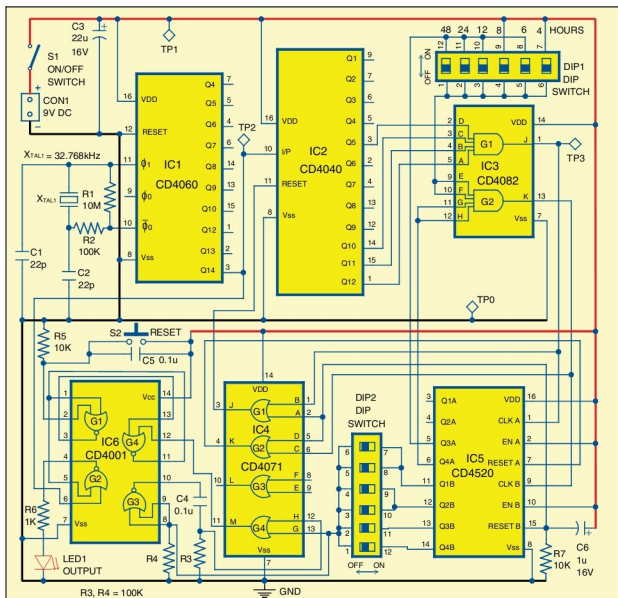


Fig. 1: Circuit diagram of the medicine-time reminder

dual 4-input AND gate CD4082 (IC3), quad 2-input OR gate CD4071 (IC4), dual up-counter CD4520 (IC5) and quad 2-input NOR gate CD4001 (IC6).

The clock for the system is provided by IC1 wired as an oscillator. This frequency is divided by 16384 (2^{14}) by the internal flip-flop chain of IC1 and a 2Hz stable clock frequency is available at its pin 3. Counter IC2 and 4-input AND gate G1 of IC3 are wired to divide the 2Hz clock by 3600. So, a pulse every 30 minutes is available at pin 1 of IC5.

Further division by IC5 is controlled by G2 of IC3 and the positions of DIP1 and